ELLab

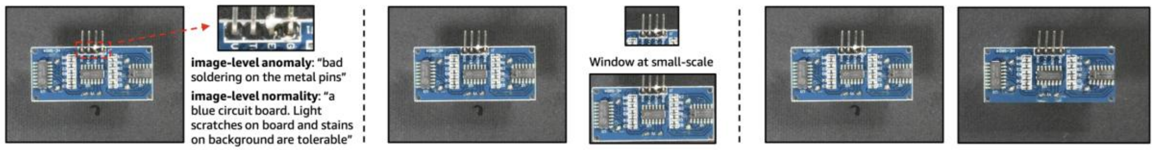
SUNGKYUNKWAN UNIVERSITY

WinCLIP Zero Shot 코드구현

2026.03.19

WinCLIP

Introduction



(a) Language defines normality and anomaly
(b) Multi-scale inspection provides comprehensive view
(c) Normal image as reference context to visually define normality

Proposed Idea (WinCLIP)

- Window-based CLIP (WinCLIP): multi-scale window feature를 이용해 language-aligned spatial feature 추출
- WinCLIP+: few-normal-shot setting에서 정상 이미지 reference 정보 추가 활용
- 추가 학습 없이(zero-shot) 다양한 anomaly detection task에 적용 가능

Main Contributions

1. Prompt ensemble로 zero-shot anomaly classification 성능 향상
2. WinCLIP: multi-scale feature 기반 language-guided anomaly segmentation
3. WinCLIP+: 정상 reference 이미지를 활용한 few-normal-shot anomaly detection
4. MVTec-AD / VisA benchmark에서 SOTA 성능 달성

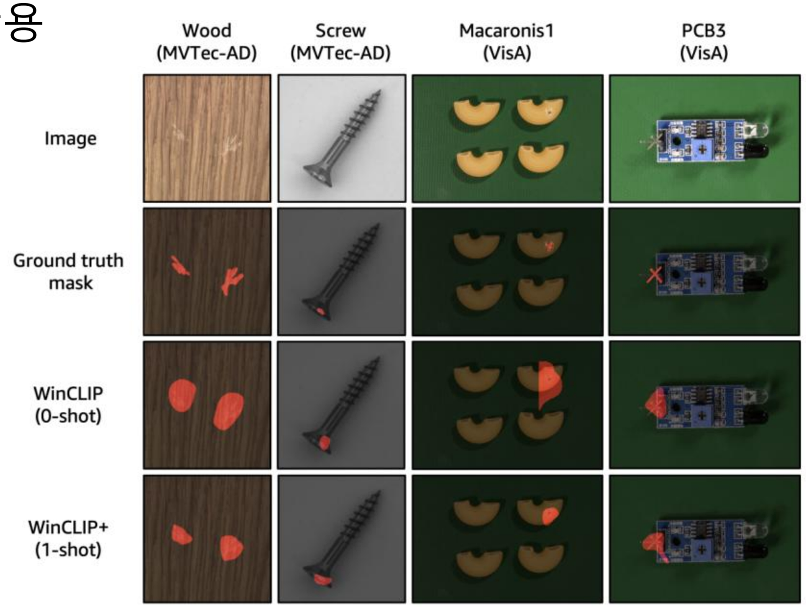


Figure 1. Language guided zero-/one-shot[1] anomaly segmentation from WinCLIP/WinCLIP+. Best viewed in color and zoom in.

WinCLIP - Language-driven zero-shot AC

Prompt $s-$: normal [object], $s+$: anomalous [object]

Two-class design $< f(x), g(s+) >, < f(x), g(s-) > \rightarrow$ softmax 계산

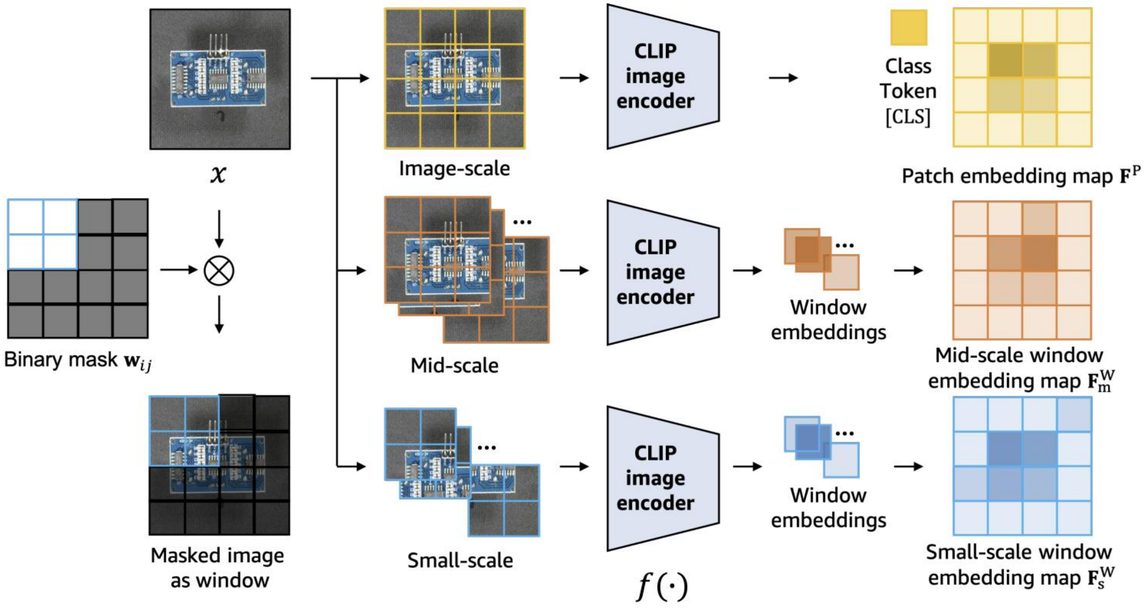
One-class design $-< f(x), g(s-) > \rightarrow$ anomaly score (normal과 멀수록 anomaly)

Compositional Prompt Ensemble

State words와 text templates를 조합하여 prompt를 생성함

flawless perfect without defect	a photo of a [c] a photo of a [c] for visual inspection a cropped photo of a [c]	normal 문장 수: $3 \times 3 = 9 \rightarrow \text{mean}$ anomalous 문장 수: $3 \times 3 = 9 \rightarrow \text{mean}$
damaged with crack with scratch	$\mathrm{c} \in \{\mathrm{c}_1, \mathrm{c}_2, \dots, \mathrm{c}_\mathrm{k}\}$ 각 object class에 대해 prototype victor 2개 총 2k개	

WinCLIP for zero-shot AS



Window-based feature extraction

CLIP은 원래 global embedding 중심 모델이기 때문에

pixel-level anomaly segmentation에 직접 사용하기 어려움

- 이미지에 sliding window를 적용
- 각 window 영역을 CLIP encoder에 입력 $\mathbf{F}^{\mathrm{W}}_{ij} := f(\mathbf{x} \odot \mathbf{w}_{ij})$

→ local feature를 추출하여 dense feature map을 생성함

Window anomaly scoring

- 각 window feature와 text embedding 사이의 similarity를 계산하여 window-level anomaly score를 계산함
- Harmonic averaging 이용함 → 작은 값에 더 민감함
- 윈도우 중 하나라도 nomal 값에 가까우면 전체 score가 작아짐 -> false positive 감소

$$\bar{\mathbf{M}}^{\mathrm{W}}_{0,ij} := \left(\frac{1}{\sum_{u,v} (\mathbf{w}_{uv})_{ij}} \sum_{u,v} \frac{(\mathbf{w}_{uv})_{ij}}{\mathbf{M}^{\mathrm{W}}_{0,uv}} \right)^{-1}$$

$w_{i,j}$ → Image Encoder → $CLS_{w_{i,j}}$

Prototype vector

→ similarity 계산 → $M_{i,j}$ anomaly score 계산

→ 윈도우별로 harmonic averaging & multi-scale 처리

코드 구현

WinCLIP Zero Shot

objects	pixel_AUC	image_AUC
carpet	81.8	99.9
bottle	70.6	99.6
hazelnut	88.4	92.3
leather	71.7	99.6
cable	79	88.2
capsule	82.9	69.3
grid	72.2	97.7
pill	61	76.5
transistor	61.8	65.9
metal_nut	62.4	92.1
screw	80.6	73.5
toothbrush	74.3	86.9
zipper	55	78.3
tile	70.1	97.9
wood	80.5	97.4
mean	72.8	87.7

- Image 단위 정확도는 전반적으로 모든 class에서 꽤 높음
- pixel 단위 정확도는 몇몇 class에서 낮음

https://colab.research.google.com/drive/1nN-FtSZhwq7CkWAHqfw5d4P5NPxmAd0l?usp=sharing

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